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TITLE : Dementia Care System: Python-based Voice alarms and prompts , Geo-Fencing

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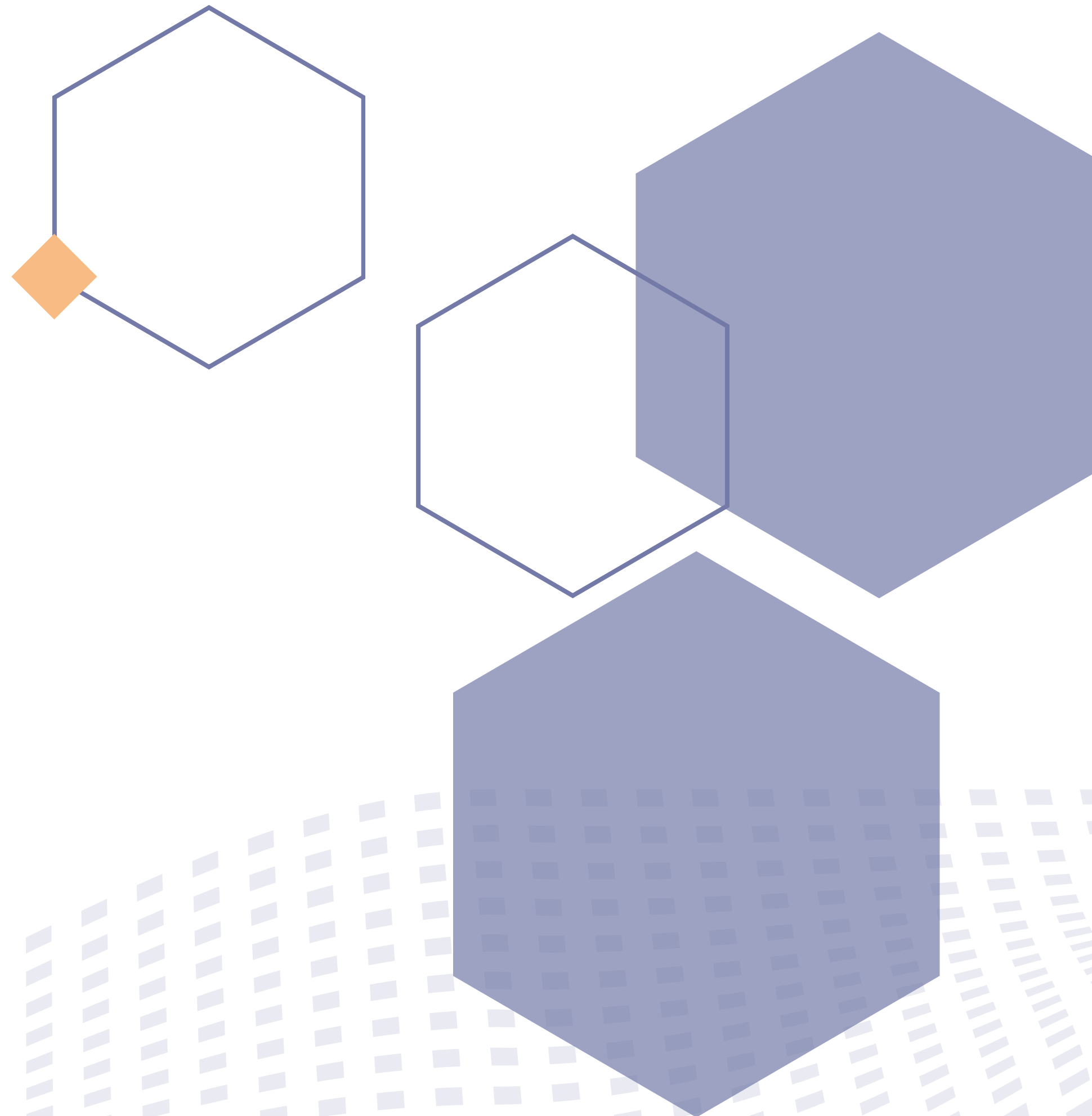
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ABSTRACT

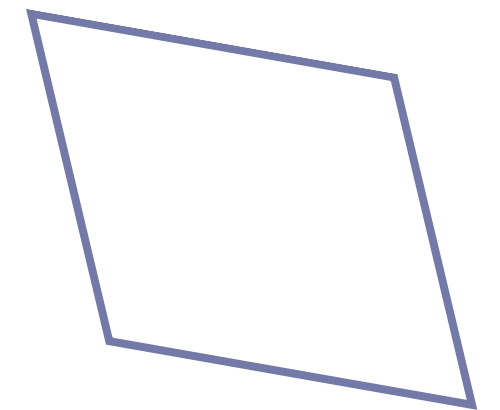
Remembering daily tasks can be challenging for many people, but this issue is often managed using task managers and alarms. However, individuals suffering from dementia require special care and attention to adhere to their routines. Traditional alarm systems often have low effectiveness for dementia patients. This project addresses the challenges dementia patients face in following daily routines and aims to reduce the stress experienced by caregivers. The software solution incorporates features such as persistent voice recognition alarms, GPS-based location tracking, geofencing and voice integration. The voice-activated alarm system will continuously prompt the patient until a response is received, ensuring task completion and notifying the caregiver. By offering personalized scheduling and voice-activated alerts for important appointments, reminders, and chores, the system serves as a helpful tool for caregivers. In addition to reducing caregiver burden, this solution empowers patients, helping them feel more confident and independent.

PROBLEM STATEMENT

Remembering daily tasks can be challenging for many people, but this issue is often managed using task managers and alarms. However, individuals suffering from dementia require special care and attention to adhere to their routines.

INTRODUCTION

- Persistent voice prompts , reminders ensure the completion of the task and notifying the caregiver . This system aims on reducing the anxiety and stress caused due to managing tasks and daily activites of dementia patients.
- The proposed system can be made adaptable to different types of Dementia as it will provide gentle repetitive reminders and use location tracking feature. Personalized task scheduling according to patients needs makes it individual friendly.
- This provides tailored interventions that match each patients unique condition and stage of dementia



RESEARCH GAP

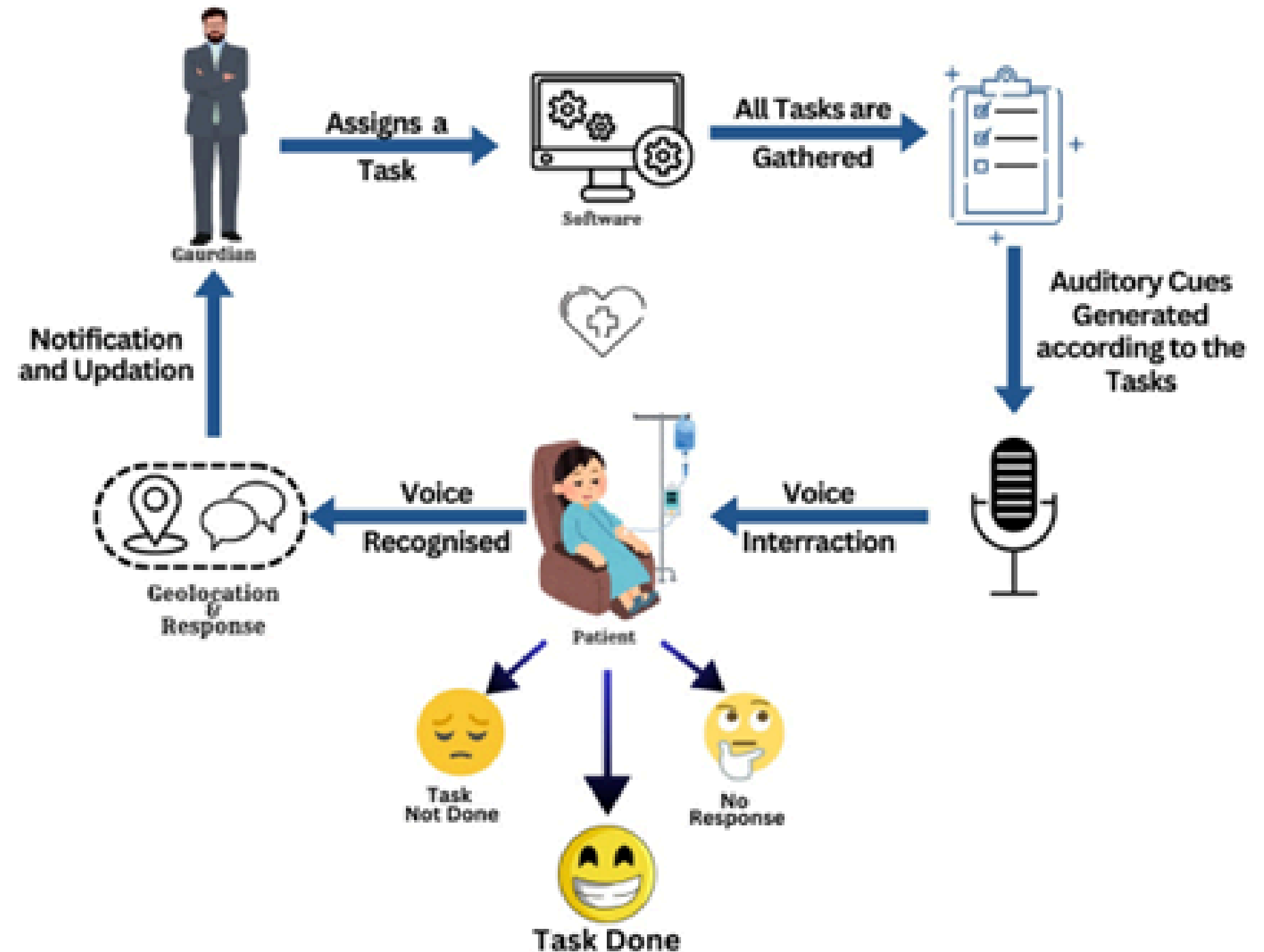
Lack of integrated systems combining reminders, voice interaction, and geolocation.

- . Current tools are not dementia-specific or accessible for elderly users.
- . No persistent, voice-based confirmation mechanisms.
- . Caregiver stress is largely unaddressed by existing tech.

METHODOLOGY

System Features

- Task Scheduling: Caregivers set tasks → voice reminders
- Voice Prompts: Repeats until patient replies
- Speech Recognition: Captures "yes"/"done" as confirmation
- Guardian Alerts: Emails for missed tasks/emergencies
- Geofencing: GPS alerts if patient exits safe zone



RESULTS

Task Management

sleep
eat

+ Add Task

✕ Remove Task

✎ Update Task

✓ Finish Adding Tasks

📍 Geofence Tracker

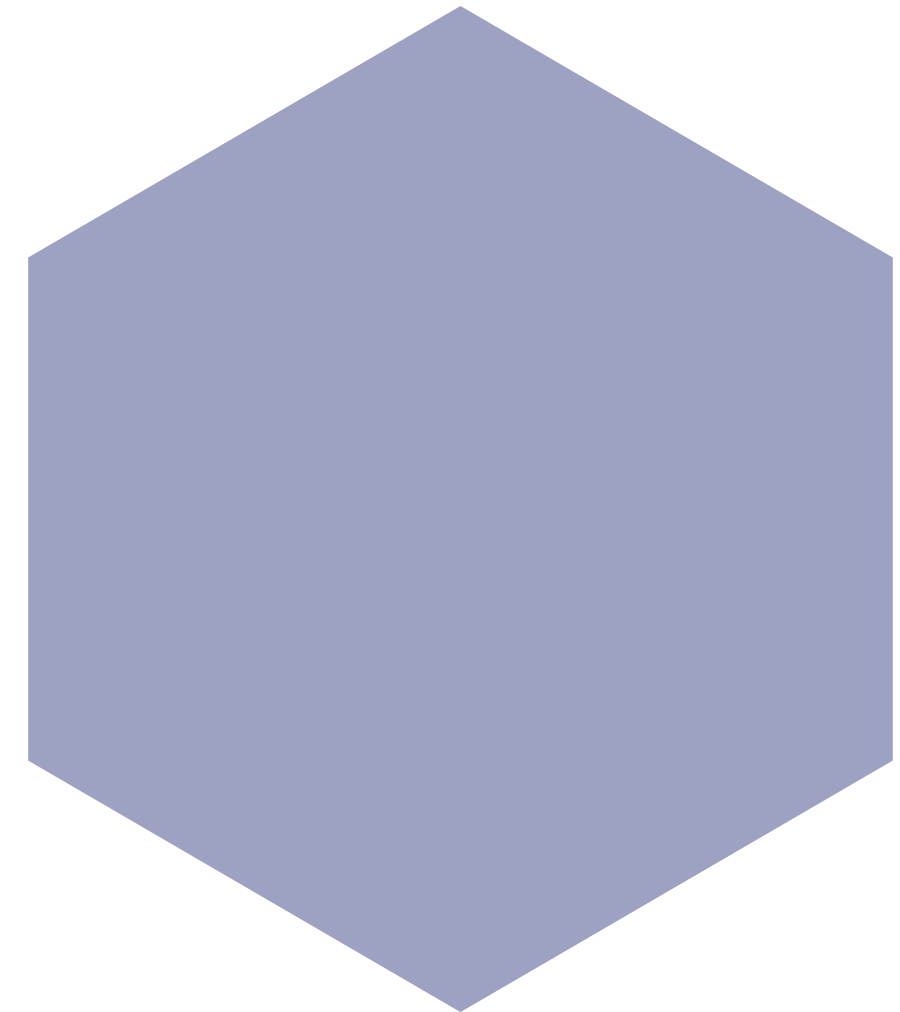
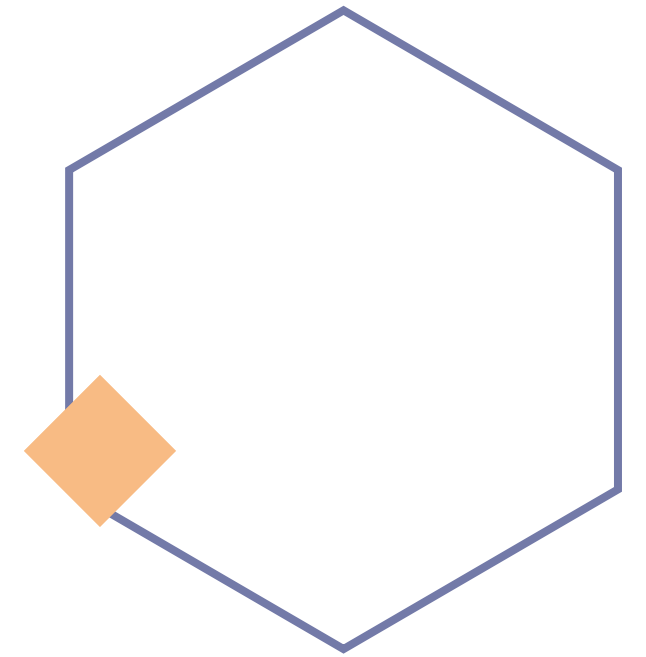
No alarm set

CONCLUSION

- Promotes independence and reduces caregiver stress
- Ensures task completion via persistent alarms
- Notifies caregivers proactively with real-time alerts

FutureScope

- Integration with smartwatches and wearables
- Machine learning for behavioral personalization
- Multilingual support and smart home integration
- Emotional detection and emergency response features



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